accelerator the size of the whole universe. (The energy required to probe matter at the Planck length is roughly equal to a thousand kilowatt-hours—the energy needed to run an average air conditioner for about one hundred hours—and so is not particularly outlandish. The seemingly insurmountable technological challenge is to focus all of this energy on a single particle, that is, on a single string.) As the U.S. Congress ultimately canceled funding for the Superconducting Supercollider—an accelerator a "mere" 54 miles in circumference—don't hold your breath while waiting for the money for a Planck-probing accelerator. If we are going to test string theory experimentally, it will have to be in an indirect manner. We will have to determine physical implications of string theory that can be observed on length scales that are far larger than the size of a string itself.15

In their groundbreaking paper, Candelas, Horowitz, Strominger, and Witten took the first steps toward this goal. They not only found that the extra dimensions in string theory must be curled up into a Calabi-Yau shape, but they also worked out some of the implications this has on the possible patterns of string vibrations. One central result they found highlights the amazingly unexpected solutions string theory offers to longstanding particle-physics problems.

Recall that the elementary particles that physicists have found fall into three families of identical organization, with the particles in each successive family being increasingly massive. The puzzling question for which there was no insight prior to string theory is, Why families and why three? Here is string theory's proposal. A typical Calabi-Yau shape contains holes that are analogous to those found at the center of a phonograph record, or a doughnut, or a "multidoughnut", as shown in Figure 9.1. In the higher-dimensional Calabi-Yau context, there are actually a variety of different types of holes that can arise—holes which themselves can have a variety of dimensions ("multidimensional holes")—but Figure 9.1 conveys the basic idea. Candelas, Horowitz, Strominger, and Witten closely examined the effect that these holes have on the possible patterns of string vibration, and here is what they found.

There is a family of lowest-energy string vibrations associated with each hole in the Calabi-Yau portion of space. Because the familiar